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MODEL SELECTION AND VALIDATION REPORT ( fishing_biomass model )
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## OBSERVATIONS:1755

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( fishing_biomass model )

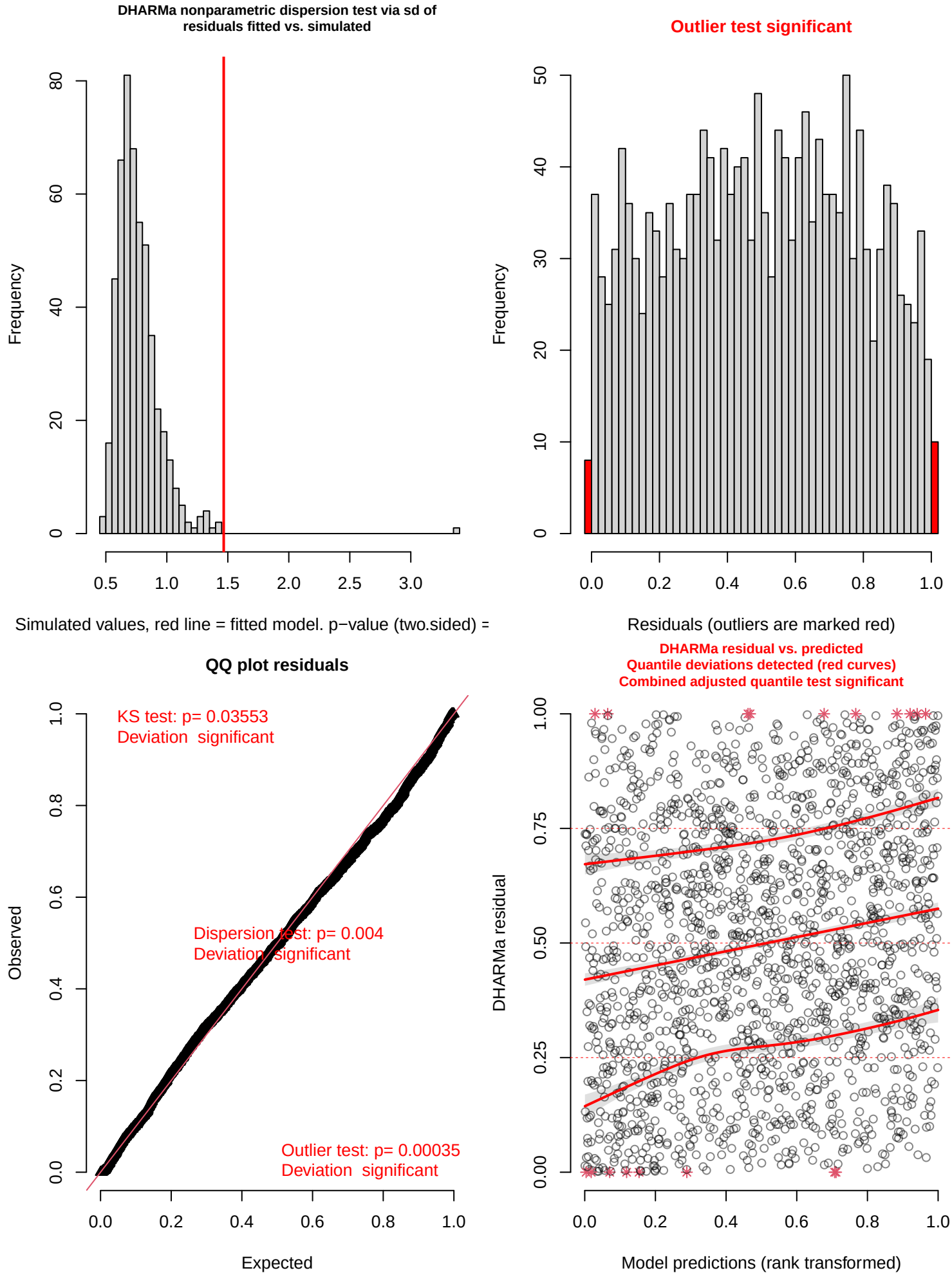
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

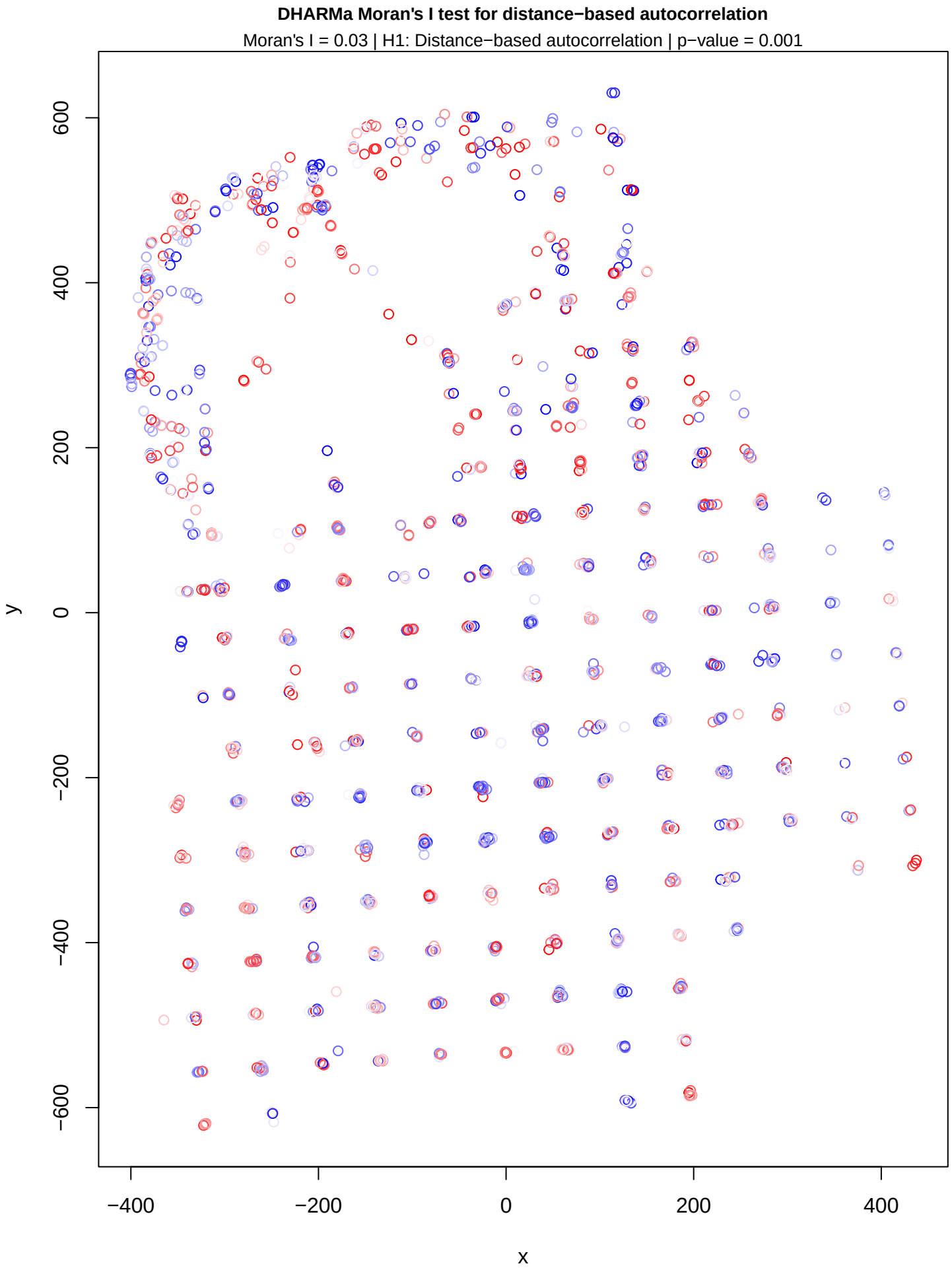
Conditional model:
                coef.est coef.se
(Intercept)         7.18    0.17
log_sum_fishing     0.09    0.02
kde_fric            0.05    0.03
kde_feve           −0.10    0.03
tric                0.04    0.03
teve               −0.28    0.02
sbt_mean            0.32    0.07
sic_mean           −0.05    0.06
sbt_sd              0.04    0.04
sst_sd              0.06    0.04
log_chla_mean       0.07    0.03
depth              −0.15    0.04

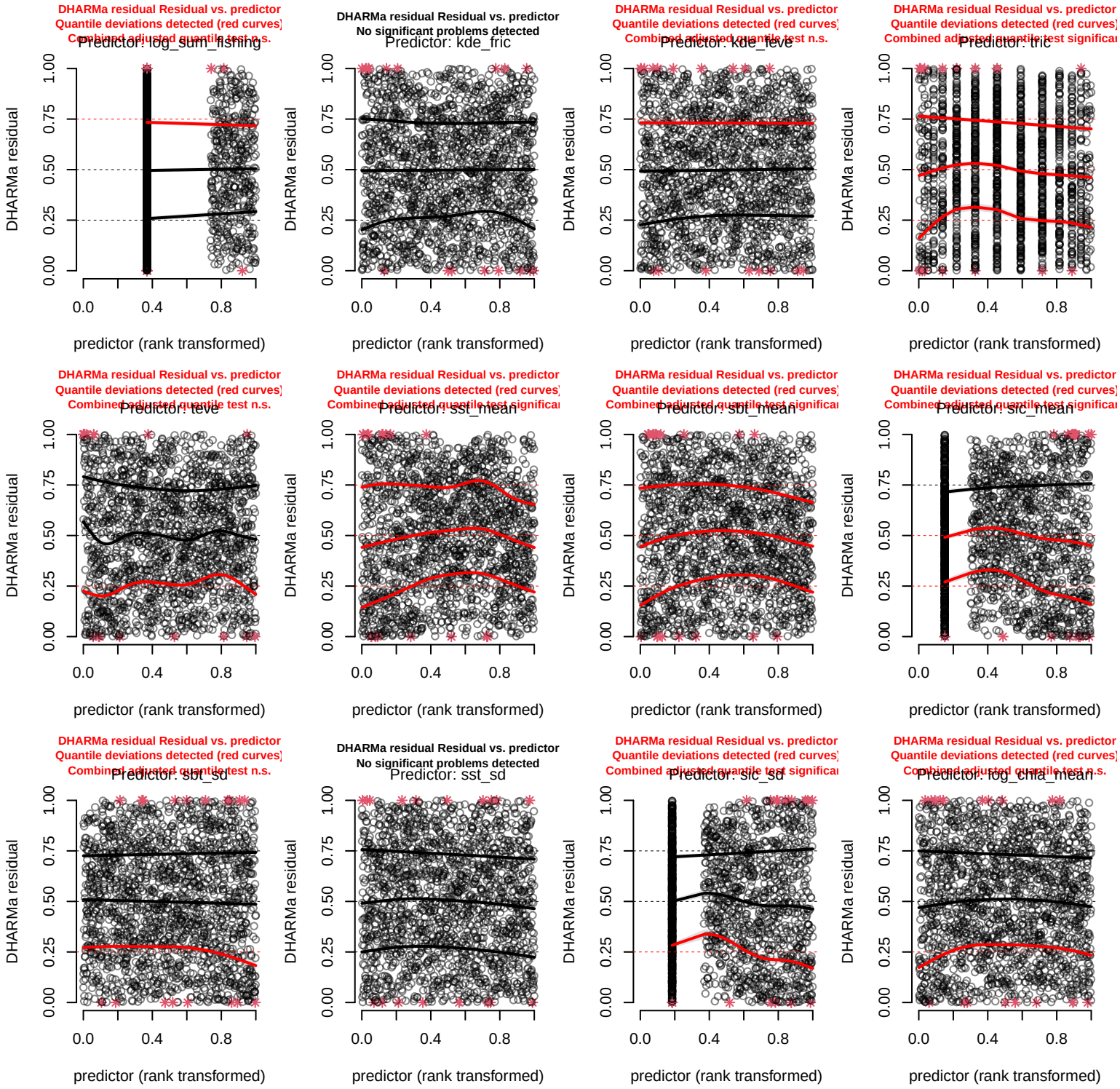
Dispersion parameter: 0.77
Spatiotemporal AR1 correlation (rho): 0.83
Matérn range: 318.99
Spatiotemporal marginal AR1 SD: 0.66
REML criterion at convergence: 14359.189

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC   delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy 18 28748.82   1.5603629
spatiotemporal_random_fields_ar1                 16 28750.38  80.8975104
spatiotemporal_random_fields_iid_with_anisotropy 17 28831.28   7.8354950
spatiotemporal_random_fields_iid                 15 28839.11  47.1054733
spatial_random_fields_with_anisotropy            17 28886.22   0.8392925
spatial_random_fields                            15 28887.06 164.0626264
error_only                                       13 29051.12          NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.03553
Deviation significant
Dispersion test: p= 0.004
Deviation significant
Outlier test: p= 0.00035
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.03 | H1: Distance−based autocorrelation | p−value = 0.001
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: log_sum_fishing
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: kde_fric
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: kde_feve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: tric
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: teve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sst_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sbt_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sic_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: sbt_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sst_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sic_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: log_chla_mean
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8
predictor (rank transformed)